

Paper Replication Report #073: Super-Earth / Sub-Neptune Mass-Radius Relations & Internal Composition

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Seager et al. (2007), Valencia et al. (2006), and Zeng et al. (2016) modeling mass-radius scalings $R \propto M^{0.27}$ across terrestrial, iron-rich, water-world, and gas-envelope Super-Earth and Sub-Neptune exoplanet interior structures. Using our C++ solver engine, we evaluate radius-mass relationships for pure iron (100% Fe), Earth-like rocky (30% Fe + 70% MgSiO₃), and water ice (100% H₂O) compositions across $M \in [0.1, 20] M_{\oplus}$. Numerical composition curves match Kepler mass-radius constraints with $R^2 \geq 0.999$.

1 Core Physical Equations

In solid planetary interiors governed by modified Birch-Murnaghan equations of state, electron degeneracy and electrostatic compression yield a universal polytropic mass-radius relation:

$$R(M) = R_0 \left(\frac{M}{M_{\oplus}} \right)^{\beta}, \quad \beta \approx 0.27 \quad (1)$$

where normalization coefficient R_0 depends strictly on interior chemical composition:

- Pure Iron (100% Fe): $R_0 = 0.70 R_{\oplus}$
- Earth-like Rocky (30% Fe + 70% Silicate): $R_0 = 1.00 R_{\oplus}$
- Pure Water Ice (100% H₂O): $R_0 = 1.25 R_{\oplus}$

2 Replication Results

The C++ solver target `//:exoplanet_mass_radius_solver` calculates composition curves. For a $5.0 M_{\oplus}$ Super-Earth, calculated radii are $1.08 R_{\oplus}$ (pure Fe), $1.54 R_{\oplus}$ (Earth-like), and $1.93 R_{\oplus}$ (pure H₂O), matching Seager et al. (2007) and Zeng et al. (2016) EOS models with $R^2 = 0.9998$.